

Reading Support: Describing Admissible Strings II

Companion to Week 7

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LING487 · Fall 2026

Regular expressions arise from a recursive definition

Supports: Regular expressions are recursively defined descriptions

For alphabet Σ , the empty language, the empty string, and each alphabet symbol are basic regular expressions. If u and v are regular expressions, union $u \cup v$, concatenation uv , and repetition u^ are also regular expressions. Nothing else enters the class.*

Each expression denotes a language, so the constructors operate on stringsets through their descriptions.

MOLR §4.3

Union, concatenation, and star preserve different information

Supports: Three operations build every regular language

Union admits strings from either described language. Concatenation places a string from the first before a string from the second. Kleene star admits any finite number of concatenations, including zero, which contributes ϵ .

Parentheses matter because they determine which subexpression each operation takes as its argument.

MOLR §4.3

Equivalent expressions need not be equally revealing

Supports: Equivalent expressions can differ greatly in intelligibility

Regular-expression identities allow several descriptions of the same language. For example, $(u \cup v)^$, $(u^* \cup v)^*$, and $u^*(vu^*)^*$ admit the same mixtures of u and v .*

Formal equivalence preserves the stringset. Expository simplicity depends on which regularity the analysis needs readers to see.

MOLR §4.3

A positive expression can exclude a forbidden substring

Supports: A regexp can state a negative local condition positively

The language over $\{0, 1, 2\}$ with no substring 01 can be described by arranging the alternatives so that any nonfinal run of zeros meets a 2, while initial 1's and 2's remain unrestricted.

The expression need not contain a negation sign. Its permitted concatenations can make the forbidden adjacency unreachable.

MOLR §4.3

A linear grammar has one continuing nonterminal

Supports: A linear grammar carries only one nonterminal at a time

In a linear grammar, every right-hand side contains at most one nonterminal. A rule may place terminal material on both sides of it, as in $A \rightarrow uBv$, or terminate the dependency with $A \rightarrow w$.

The grammar can coordinate two growing regions without creating two independently continuing context-free branches.

MOLR §4.4

Palindromes show what two-sided growth preserves

Supports: Linear does not mean merely left-to-right

Rules such as $S \rightarrow aSa \mid bSb \mid a \mid b \mid \epsilon$ make the same choice at both ends of the remaining nonterminal. The resulting yield reads the same from left to right and right to left.

Linearity concerns the number of continuing nonterminals, not the direction in which terminals appear.

MOLR §4.4

Regular relations pair two strings

Supports: A regular relation describes aligned input-output pairs

A regular relation is a subset of $\Sigma^ \times \Gamma^*$ built from paired symbols, empty elements, union, concatenation, and star. A pair may align a symbol with another symbol, with ϵ , or align ϵ with an output symbol.*

The relation becomes a function only if every admitted input has exactly one output.

MOLR §4.4

Rewrite rules abbreviate sets of correspondences

Supports: Rewrite notation can hide a relation

The familiar notation $A \rightarrow B/C_D$ describes input-output correspondences in which A in the stated environment corresponds to B . It is not a single event but a compact description of many paired strings.

Whether the relation is regular depends on the descriptive resources permitted in the context and on the representation of the input and output alphabets.

MOLR §4.4